

Zero Trust Architecture for Enterprise Security

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A hands-on simulation exploring modern security approaches



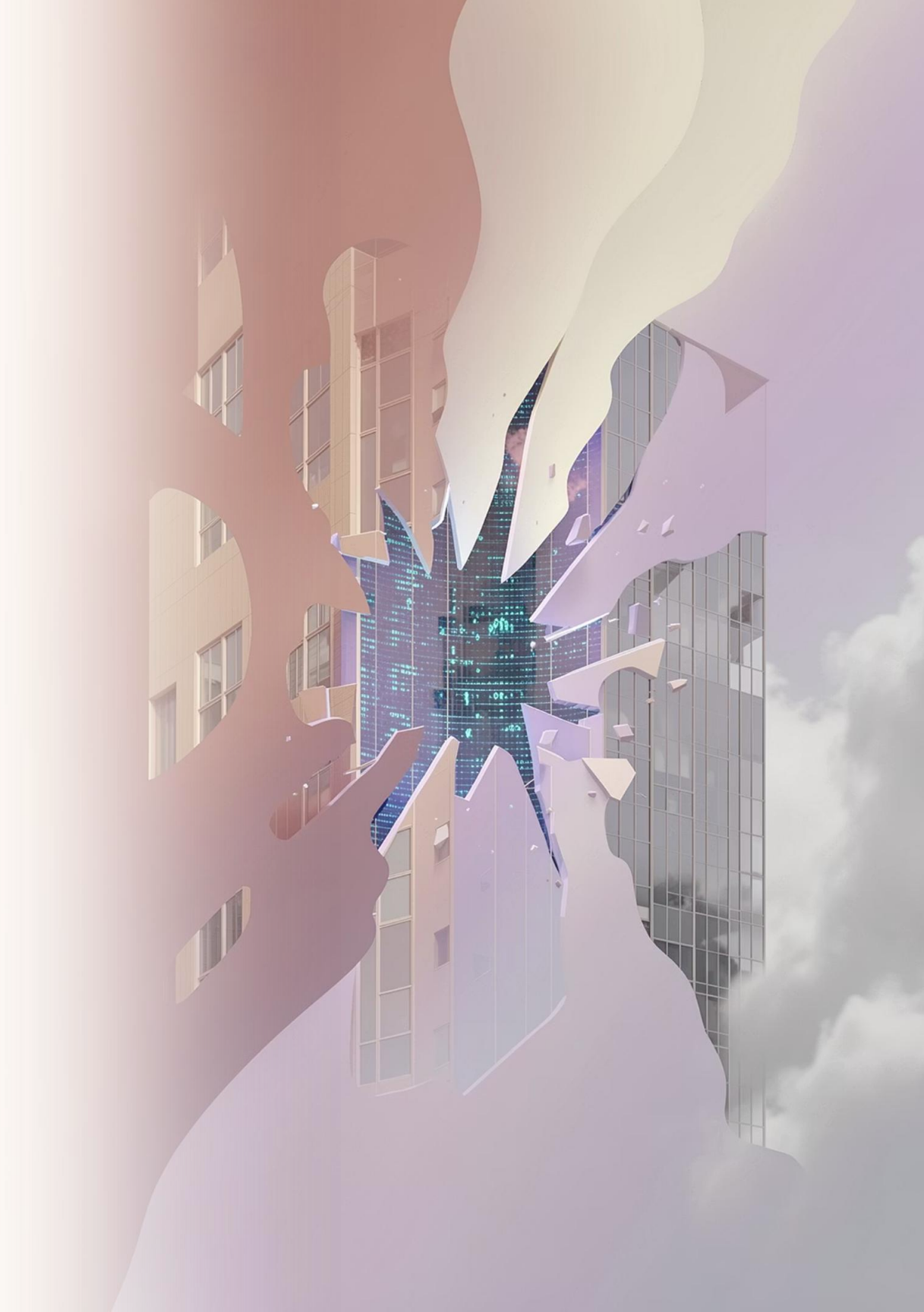
The Old Model No Longer Works

Traditional Perimeter Security

Companies used to build strong walls around office networks. Inside the building? You're trusted. Connected to Wi-Fi? You're safe.

Why It Failed

Remote work, cloud services, and sophisticated hackers made the "office walls" disappear. The perimeter model can't protect against modern threats.



What is Zero Trust?



Never Trust by Default

Every connection request must be verified, even from inside the network.



Assume Breach

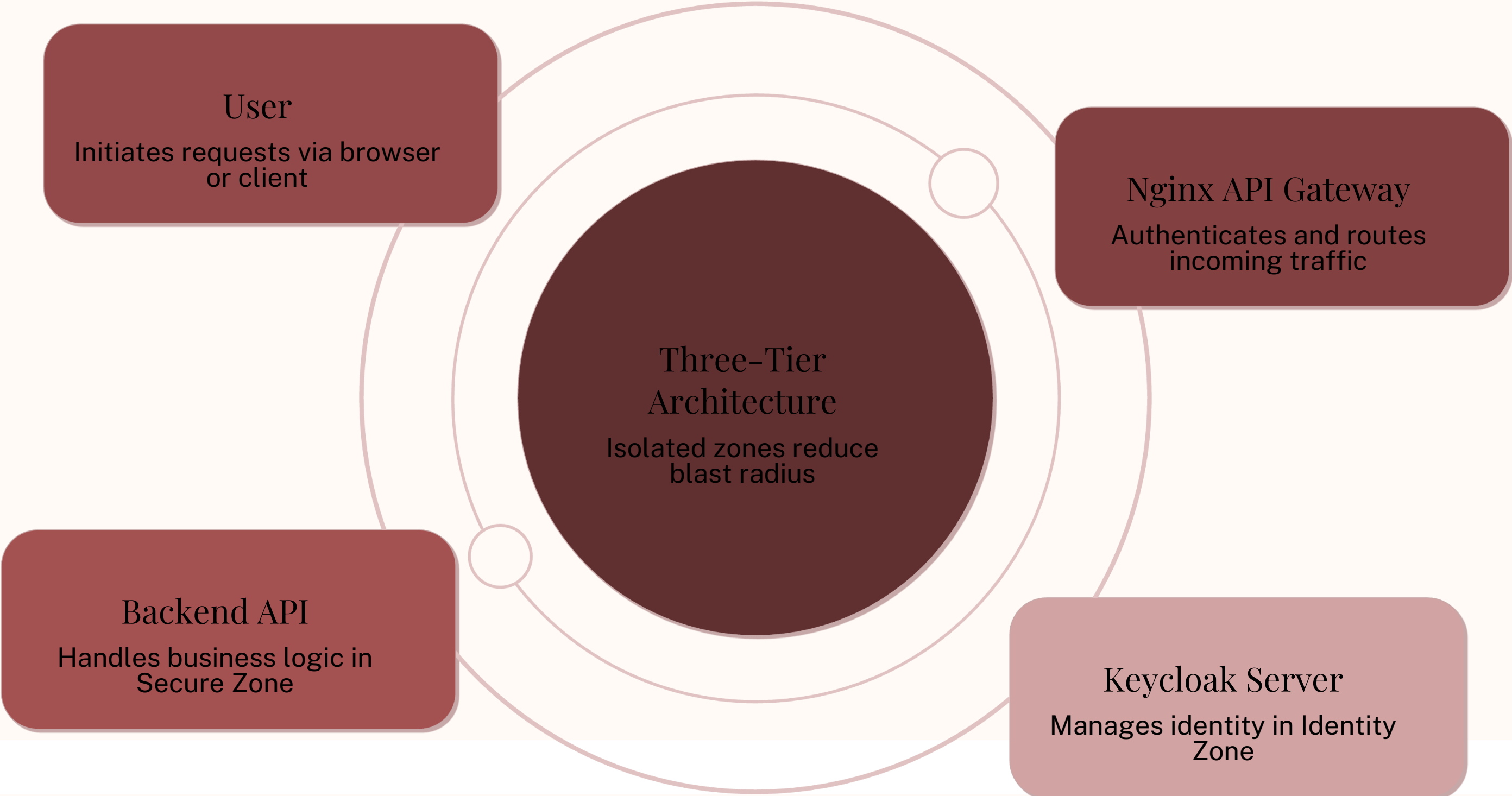
Treat the network as if it's already compromised, forcing tighter security at every level.



Verify Every Time

Each request to access files or applications gets checked before approval.

Project Architecture



Gateway Routing Rules

Request	Service	Security Reason
/api/*	zta-secure-api	Restricts API access to verified users only
/auth/*	zta-iam-keycloak	Centralized path for login and identity checks
/*	Blocked (403)	Blocks any unauthorized or unknown paths



Implementation Methodology

01

Network Segmentation

Docker bridge networks isolate backend components. API hidden on private network, accessible only through Nginx gateway.

02

Identity & Access Control

Keycloak handles login with MFA enabled. Gateway validates JWT tokens before allowing traffic to API.

03

Access Policies

Role-Based Access Control with Admin and User roles. MFA required for all logins via TOTP.

Key Findings

Security Under Pressure

Direct API access attempts impossible due to network rules.
Gateway acts as mandatory checkpoint.

MFA Protection

Stolen password simulation stopped by MFA challenge. Second factor effectively blocks unauthorized access.

Isolation Benefits

Gateway issues left backend API completely hidden. Network segmentation provides resilience.



Project Observations



Location is Irrelevant

You can't trust anyone just because they're in the office. Physical location doesn't equal trustworthiness.



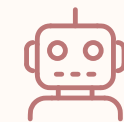
Identity is Everything

Strengthening identity verification is the best way to secure the system against modern threats.



Simple Rules, Strong Security

Clear gateway rules make security management easier and more reliable in practice.



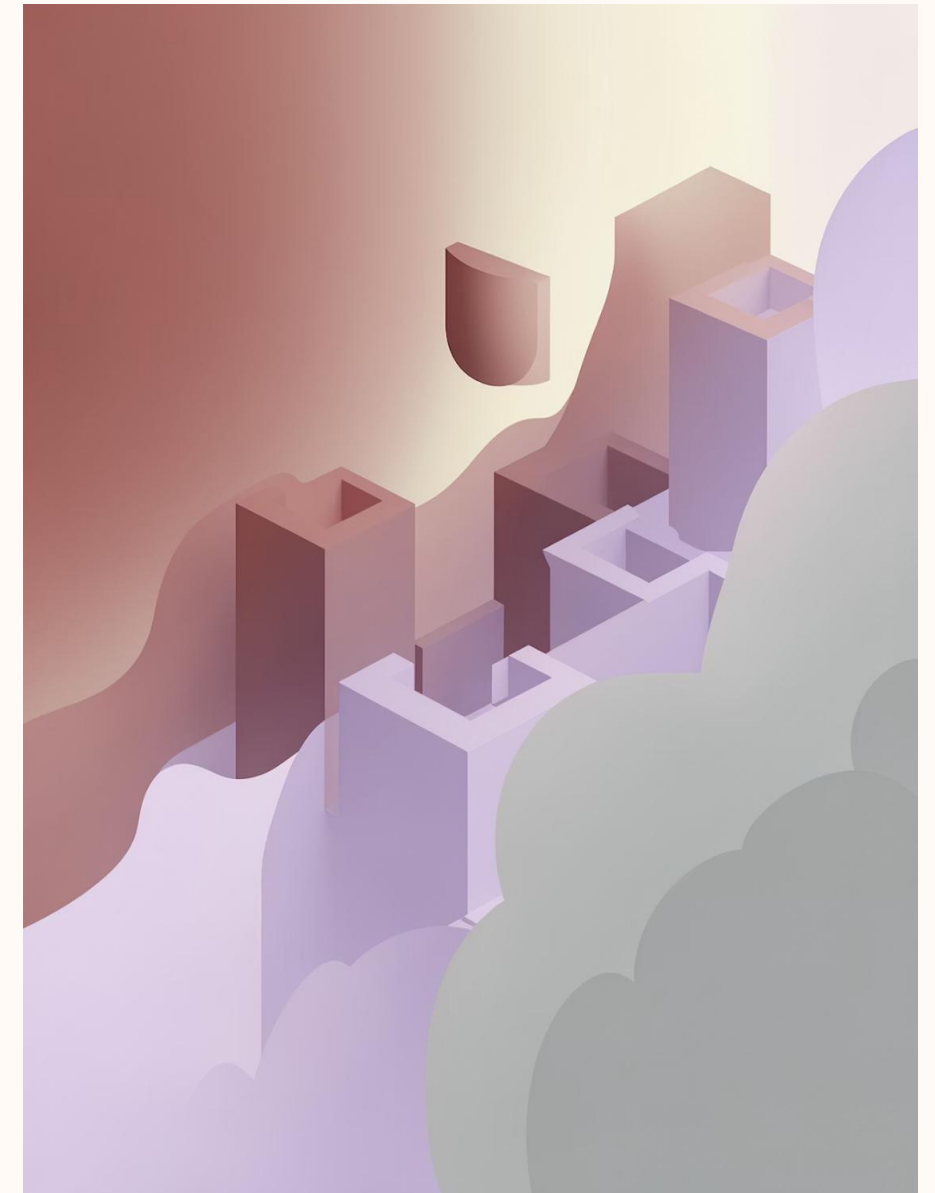
Automation Helps

Docker makes deploying and testing security zones much faster and more consistent.

Conclusion

Zero Trust Architecture proves highly effective against modern security challenges. By abandoning the old perimeter model and focusing on identity verification and network isolation, we can build systems that are significantly harder for attackers to compromise.

This simulation demonstrated practical implementation using Docker, Nginx, and Keycloak — showing how real-world security can be achieved with accessible tools and clear policies.



Technical Implementation

Requirements

- Docker Desktop or Docker Engine
- Docker Compose for orchestration

Project Files

- `docker-compose.yml` - Container definitions
- `nginx.conf` - Gateway routing rules

Service Configuration

- Keycloak server on port 8080
- Nginx gateway on port 443
- Backend API on internal network

THANK YOU

